

36) Prove or disprove that for all sets A, B and C, we have

a) $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

Solution: Let, (x, y) be an arbitrary element of $A \times (B \cup C)$, i.e. $(x, y) \in A \times (B \cup C)$.
 By Defn. 8 from Section 2.1, $x \in A \wedge y \in (B \cup C)$. By Defn. of AND, $x \in A$ and $y \in (B \cup C)$. By Defn. of ~~union~~, $y \in B \vee y \in C$.
 By Defn. of AND, $x \in A \wedge (y \in B \vee y \in C)$. By 2nd Distributive Law of Proposition, $(x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$. By Defn. 8 from Section 2.1, $(x, y) \in (A \times B) \vee (x, y) \in (A \times C)$. By Defn. 1, $(x, y) \in (A \times B) \cup (A \times C)$.

By Defn. 3 from Section 2.1, $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$.
 Let, (x, y) be an arbitrary element of $(A \times B) \cup (A \times C)$, i.e. $(x, y) \in (A \times B) \cup (A \times C)$.
 By Defn. 1, $(x, y) \in (A \times B) \vee (x, y) \in (A \times C)$. By Defn. 8 from Section 2.1, $(x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$. By 2nd Distributive Law of Proposition, $x \in A \wedge (y \in B \vee y \in C)$. By Defn. 1, $x \in A \wedge y \in (B \cup C)$. By Defn. 8 from Section 2.1, $(x, y) \in A \times (B \cup C)$.
 By Defn. 3 from Section 2.1, $(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$.

$\therefore A \times (B \cup C) = (A \times B) \cup (A \times C) \quad \square$

b) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

Solution: Let, (x, y) be an arbitrary element of $A \times (B \cap C)$, i.e. $(x, y) \in A \times (B \cap C)$.
 By Defn. 8 from Section 2.1, $x \in A \wedge y \in (B \cap C)$. By Defn. of AND, $x \in A$ and $y \in (B \cap C)$. By Defn. 2, $y \in B \wedge y \in C$. By Defn. of AND, $x \in A$ and $y \in B$ and $y \in C$.
 By Defn. of AND, $(x \in A \wedge y \in B) \wedge (x \in A \wedge y \in C)$. By Defn. 8 from Section 2.1, $x \in A \times B$ and $x \in A \times C$. By Defn. 3 from Section 2.1, $(x, y) \in (A \times B) \cap (A \times C)$. By Defn. 2, $A \times (B \cap C) \subseteq (A \times B) \cap (A \times C)$.
 Let, (x, y) be an arbitrary element of $(A \times B) \cap (A \times C)$, i.e. $(x, y) \in (A \times B) \cap (A \times C)$.
 By Defn. 2, $(x, y) \in (A \times B) \wedge (x, y) \in (A \times C)$. By Defn. 8 from Section 2.1, $(x \in A \wedge y \in B) \wedge (x \in A \wedge y \in C)$. By 2nd Associative Law of